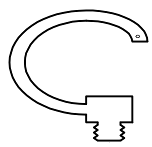
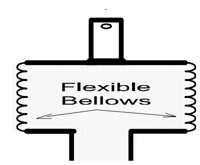
REVIEW QUESTIONS - EQUIPMENT

Note

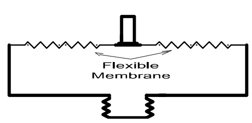
1. Briefly explain how each of the following devices is used to measure pressure.



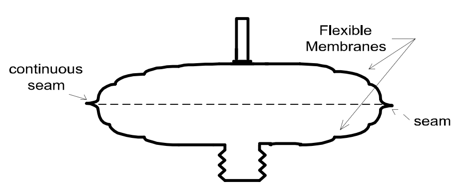
a. Bourdon Tube



b. Bellows



c. Diaphragm



d. Capsule

Note

2. Explain how the capacitor capsule, differential pressure transmitter detects pressure differences.

3. Describe a strain gauge.

4. A pressure-measuring instrument is designed around a bourdon tube. Explain how extreme changes in the ambient temperature of the bourdon tube will introduce errors into the readings of the instrument.

5. Briefly describe how the ambient pressure in a room containing a pressure transmitter can affect the reading of the transmitter.

6. Explain how flow can be measured using an orifice plate, venturi or flow nozzle.

7. Explain how elbow taps are used to measure the flow in a steam line.

Note

8. Briefly describe the how each of the following devices is used to measure flow.

   a. Orifice plate

   b. Venturi

   c. Flow nozzle

   d. Elbow tap

   e. Annubar

   f. Pitot tube

9. For each of the following devices explain how flow measurements will be affected by

   a. Changes in fluid temperature

   b. Changes in fluid pressure

   c. Erosion

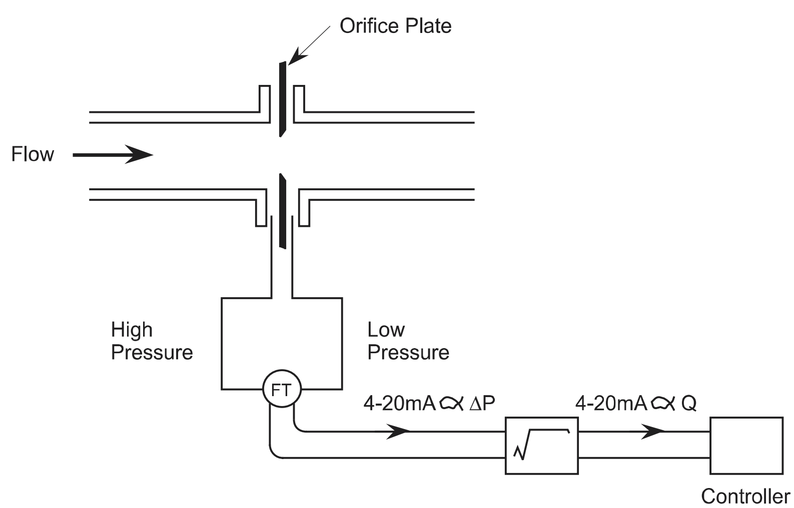
      i. Orifice plate

      ii. Venturi

      iii. Flow nozzle

      iv. Elbow tap

10. The drawing below shows a typical flow control loop. Explain the purpose of the square root extractor.



Note

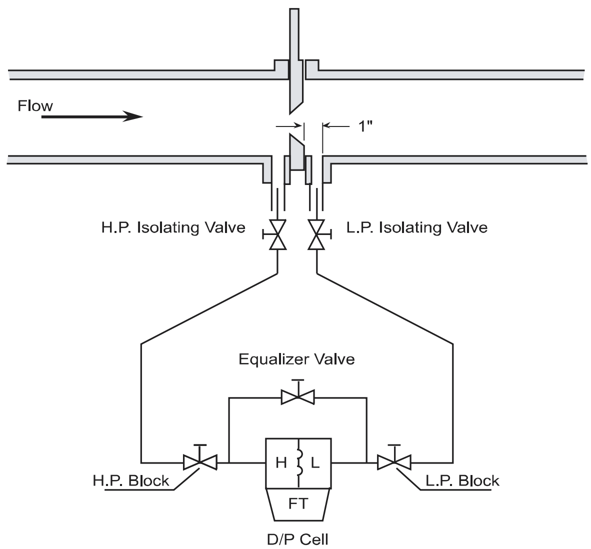
11. A differential pressure transmitter is calibrated to measure the flow of a liquid. Explain what will happen if the fluid is not pure liquid but contains some vapor bubbles.

12. On the following drawing identify the following.

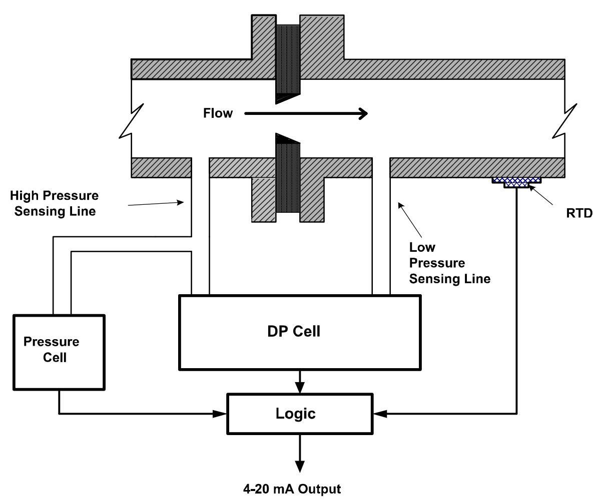
    a. Three valve manifold

    b. Primary element

    c. Transmitter



13. The following diagram shows a density corrected flow loop. Briefly explain the operation of the loop.



Note

14. In a flow loop using a venturi for a primary element what will be the consequences of the following abnormalities?

    a. Vapor formation in the throat

    b. Clogging of the throat by foreign material

    c. Leaks in the Hi pressure sensing line

    d. Leaks in the low pressure sensing line

15. Sketch the typical installation of a pressure transmitter on an open tank, measuring the level of the fluid in the tank. Explain how this transmitter derives a level signal.

16. Sketch the typical installation of a dry leg level transmitter installation. Explain how this transmitter derives a level signal.

17. Sketch the typical installation of a wet leg level transmitter installation. Explain how this transmitter derives a level signal.

18. The three-valve manifold on a level transmitter must be operated correctly when either removing the transmitter for service or returning it to service. Explain how the transmitter may be damaged by incorrect operation of the manifold.

19. Zero elevation and zero suppression are calibration techniques used in level transmitter calibrations. Explain the purpose of these techniques.

20. Explain how a bubbler is used to measure the level in an open tank.

21. Explain how a bubbler is used to measure the level in a closed tank.

22. A dry leg level transmitter installation is measuring the level of a hot water tank. What happens to the level and the level indication if the temperature of the tank is increased and no water leaves the system?

23. A dry leg level transmitter installation is measuring the level of a hot water tank. What happens to the level and the level indication if the static pressure on the tank is increased and no water leaves the system?

Note

24. Describe the effects on a level transmitter of the following abnormalities in a wet leg configuration.

    a. a leak in the high pressure sensing lines

    b. a leak in the low pressure sensing line

    c. a completed block low pressure sensing line

25. Explain how an RTD is used to measure temperature. Include in your answer a statement to explain why there are three leads from the temperature transmitter to the RTD.

26. What type of signal does a thermocouple produce?

27. RTDs are used to measure the temperature of the reactor outlet feeders. Thermo couples are used to measure temperatures on the turbine. Explain the reasons for the selection of these devices for their respective applications.

28. State the power ranges each of the following neutron detectors will provide the signal for bulk reactor power control.

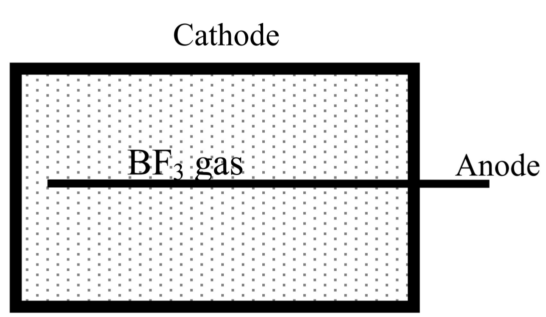
    a. Start-up instrumentation

    b. Ion Chambers

    c. In Core Detectors

29. Why is there an overlap of the ranges over which the various neutron detectors are used to control reactor power?

30. Using the following diagram explain how a neutron flux signal is derived in a BF_3 detector



Note

31. Explain the reason that BF3 detectors burnout if they are left installed when the reactor is at a high power level.

32. Briefly describe how the following ion chamber develops a signal proportional to flux.

Note

37. Explain how each of the following factors can affect the accuracy of the in-core detector flux measurement

    a. Fueling or reactivity device position

    b. Start-up of the reactor

    c. Long term exposure to neutron flux

    d. Moderator poison

38. State the ranges over which the in core detectors and the ion chambers are used to control reactor power. Explain why each is used over this range.

CONTROL

Note

3.0 INTRODUCTION

Control of the processes in the plant is an essential part of the plant operation. There must be enough water in the boilers to act as a heat sink for the reactor but there must not be water flowing out the top of the boilers towards the turbine. The level of the boiler must be kept within a certain range. The heat transport pressure is another critical parameter that must be controlled. If it is too high the system will burst, if it is too low the water will boil. Either condition impairs the ability of the heat transport system to cool the fuel.

In this section we will look at the very basics of control. We will examine the fundamental control building blocks of proportional, integral and differential and their application to some simple systems.

3.1 BASIC CONTROL PRINCIPLES

Consider a typical process control system. For a particular example let us look at an open tank, which supplies a process, say, a pump, at its output. The tank will require a supply to maintain its level (and therefore the pump's positive suction head) at a fixed predetermined point. This predetermined level is referred to as the setpoint (SP) and it is also the controlled quantity of the system.

Clearly whilst the inflow and outflow are in mass balance, the level will remain constant. Any difference in the relative flows will cause the level to vary. How can we effectively control this system to a constant level? We must first identify our variables. Obviously there could be a number of variables in any system, the two in which we are most interested are:

The controlled variable - in our example this will be level.

The manipulated variable – the inflow or outflow from the system.

If we look more closely at our sample system (Figure 1), assuming the level is at the setpoint, the inflow to the system and outflow are balanced. Obviously no control action is required whilst this status quo exists. Control action is only necessary when a difference or error exists between the setpoint and the measured level. Depending on whether this error is a positive or negative quantity, the appropriate control correction will be made in an attempt to restore the process to the setpoint.